Dijkstra

```c
#include <stdio.h>
#define V 5
#define INF 9999

void dijkstra(int g[V][V], int s) {
    int d[V], vis[V] = {0}, i, j, u, m;
    for (i = 0; i < V; i++) d[i] = INF;
    d[s] = 0;
    for (i = 0; i < V; i++) {
        for (u = -1, m = INF, j = 0; j < V; j++)
            if (!vis[j] && d[j] < m) m = d[j], u = j;
        if (u < 0) break;
        vis[u] = 1;
        for (j = 0; j < V; j++)
            if (g[u][j] && !vis[j] && d[u] + g[u][j] < d[j])
                d[j] = d[u] + g[u][j];
    }
    printf("Node\tDistance from %d\n", s);
    for (i = 0; i < V; i++) printf("%d\t%d\n", i, d[i]);
}

int main() {
    int g[V][V] = {
        {0,10,0,0,5},{0,0,1,0,2},{0,0,0,4,0},{7,0,0,0,0},{0,3,9,2,0}
    };
    dijkstra(g, 0);
    return 0;
}
```

Bellman-Ford

```c
#include <stdio.h>
#define V 5
#define E 9
#define INF 9999

typedef struct { int u, v, w; } Edge;

void bf(Edge e[], int s) {
    int d[V], i, j;
    for (i = 0; i < V; i++) d[i] = INF;
    d[s] = 0;
    for (i = 0; i < V - 1; i++)
        for (j = 0; j < E; j++)
            if (d[e[j].u] + e[j].w < d[e[j].v])
                d[e[j].v] = d[e[j].u] + e[j].w;
    for (j = 0; j < E; j++)
        if (d[e[j].u] + e[j].w < d[e[j].v])
            return (void)printf("Negative cycle detected!\n");
    printf("Node\tDistance from %d\n", s);
    for (i = 0; i < V; i++) printf("%d\t%d\n", i, d[i]);
}

int main() {
    Edge e[E] = {{0,1,10},{0,4,5},{1,2,1},{1,4,2},{2,3,4},
                 {3,0,7},{4,1,3},{4,2,9},{4,3,2}};
    bf(e, 0);
    return 0;
}
```